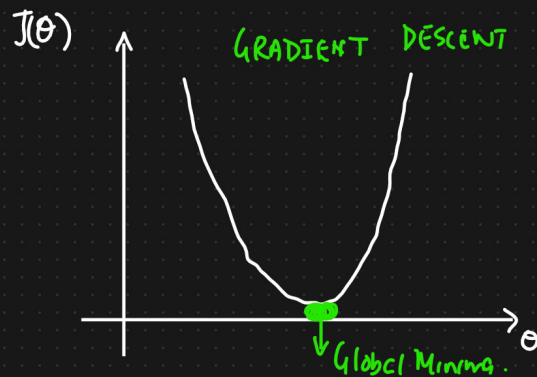
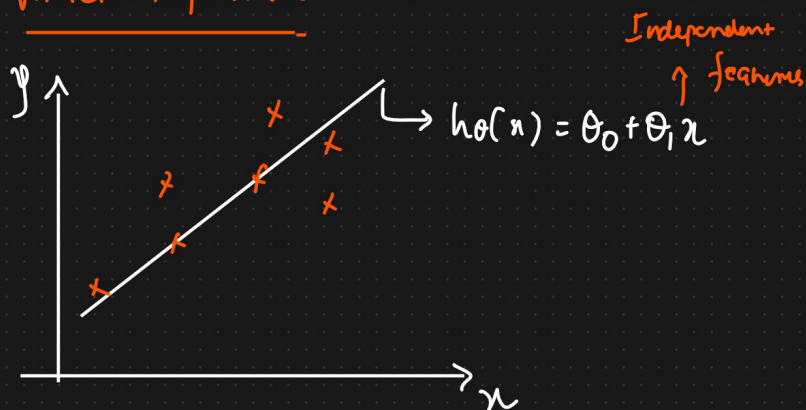


Ridge Regression, Lasso Regression, Elasticnet Regression

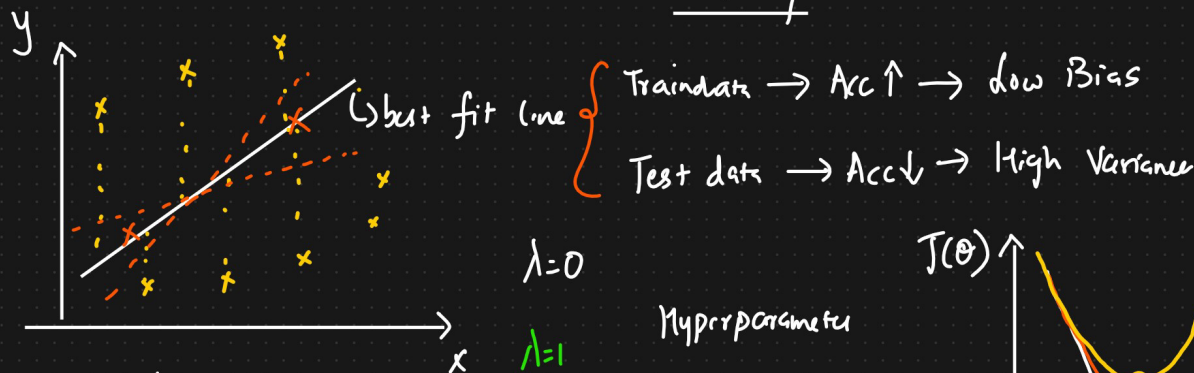
Linear Regression



$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2$$

Mean Squared Error

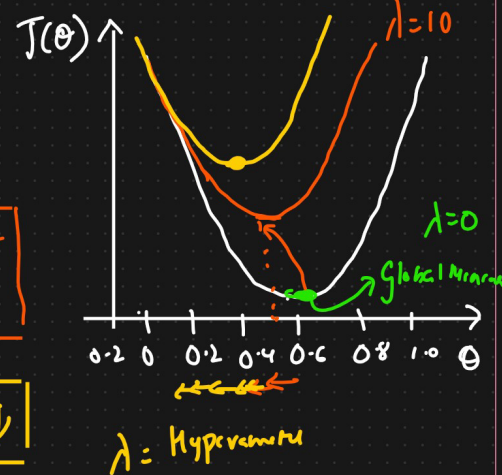
① Ridge Regression (L2 Regularization) $\rightarrow$ Reduce Overfitting



$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_0(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^m (\text{slope})^2$$

$\lambda \leq (\text{slope})^2$

$\lambda \uparrow$ slope $\downarrow$



$$\lambda = 1$$

$$> 0$$

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3$$

0.24

$0.34 + 0.52x_1 + 0.48x_2 + 0.24x_3$

$$= 0.34 + 0.40x_1 + 0.38x_2 + \boxed{0.14x_3}$$

↑↑↑

② Lasso Regression (L_1 Regularization) $\rightarrow$ Feature Selection

$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{i=1}^n |\text{slope}|$$

↑↑

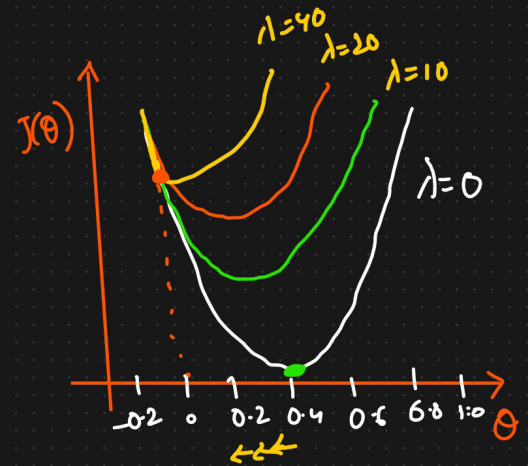
$$h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 + \theta_4 x_4$$

$$h_{\theta}(x) = 0.52 + 0.65x_1 + 0.72x_2 + 0.34x_3 + \boxed{0.12x_4}$$

$\Downarrow$
Lasso Regression

$$= 0.52 + 0.51x_1 + 0.60x_2 + 0.14x_3 + \boxed{0x_4}$$

$\nearrow 0$



③ ElasticNet Regression

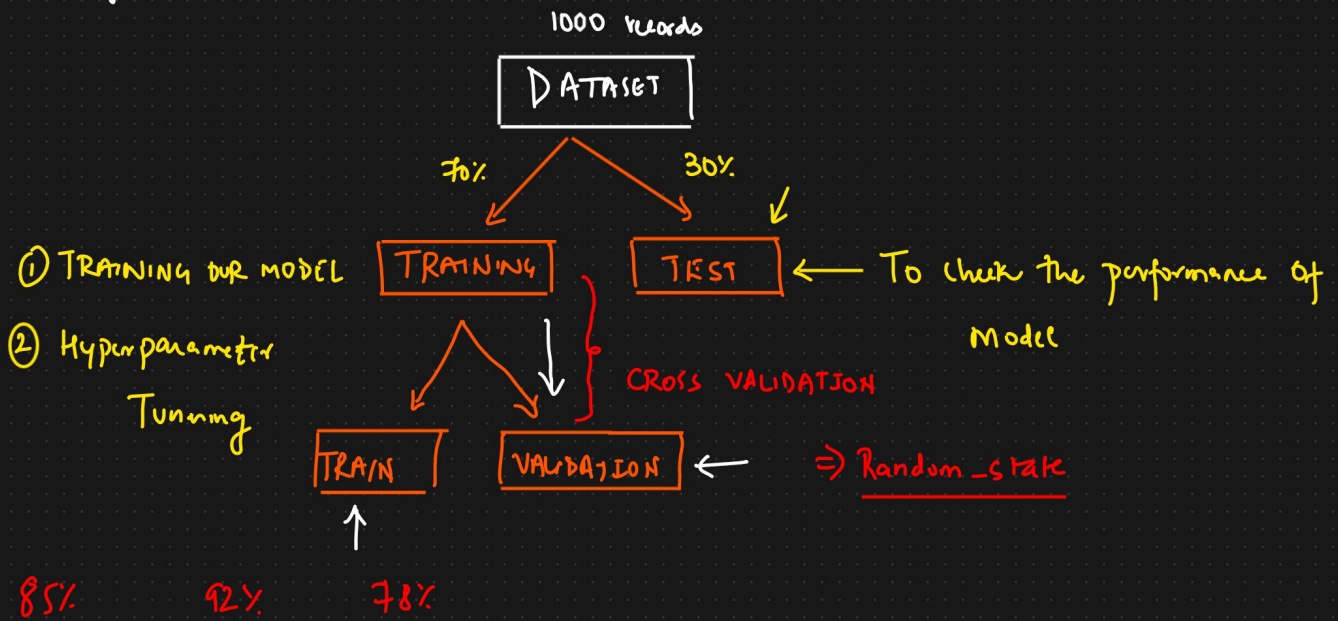
- $\rightarrow$ ① Reduce Overfitting
- $\rightarrow$ ② Feature Selection

$$\text{Cost fn} = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda_1 \sum_{i=1}^m (\text{slope})^2 + \lambda_2 \sum_{i=1}^m |\text{slope}|$$

$\Downarrow$ Reduce Overfitting $\Downarrow$ Feature Selection

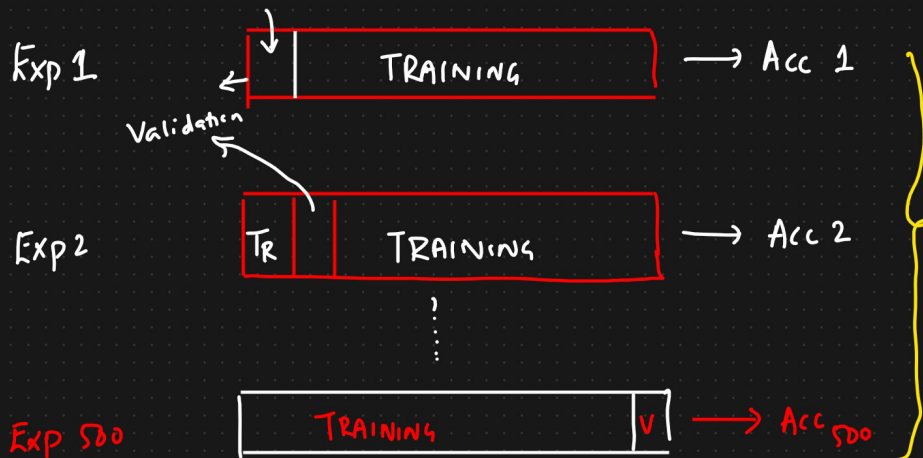
{ Hyperparameter Tuning the
Linear Regression }

Types of CROSS VALIDATION



- ① leave One Out CV (LOOCV) ② leave p out CV
- $P=10 \quad P=20 \quad P=30$

TRAINING → 500 Records ↑↑ Complexity of Training Model



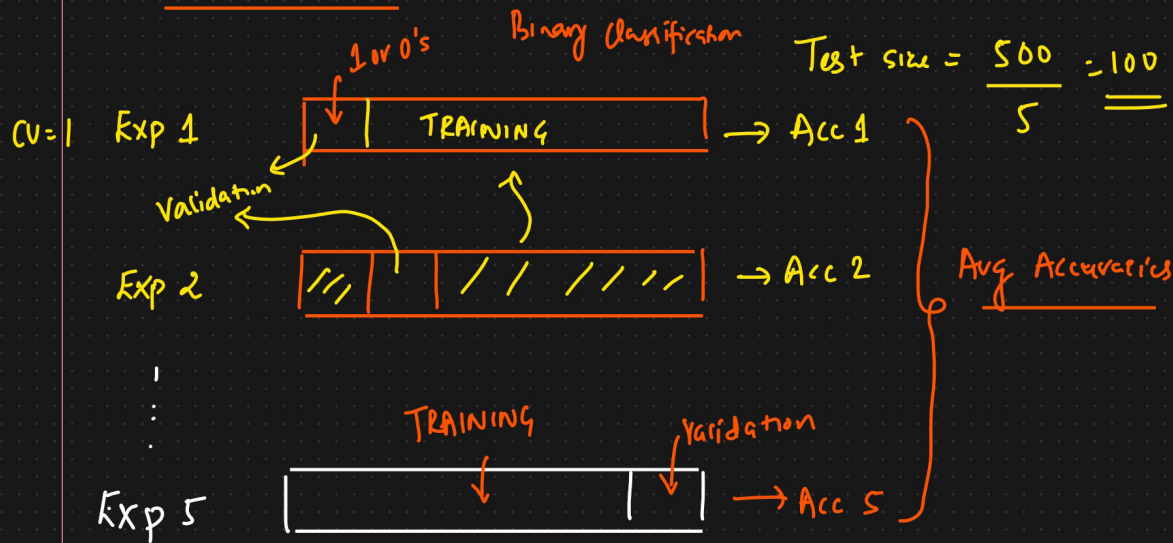
① Overfitting → TRAINING ↑↑ Acc → New Test Data → Acc ↓↓

Validation Acc ↓↓

③ K Fold CV

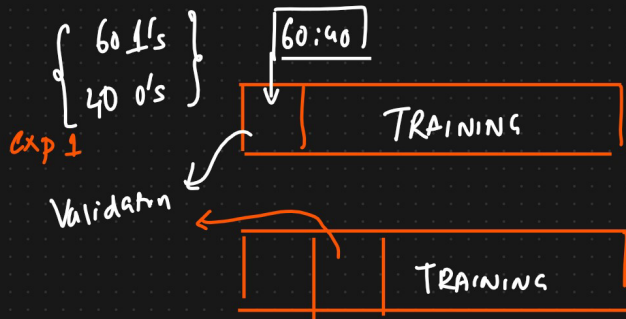
$K=5$

$n=500$



④ Stratified K Fold CV

$K=5$



Test size = $\frac{500}{5} = 100$

↓

Validation Data

⑤ Time Series CV

Reviews

Product Sentiment Analysis

Time

JAN → DEC

DAY 1 DAY 2 DAY 3 DAY 4 | DAY N

TRAINING

Validation

Time Series Application